NOTICE OF
CHANGE

DOD-STD-35-114(MI)
NOTICE 3
18 SEPTEMBER 1992

MILITARY STANDARD

AUTOMATED ENGINEERING
DOCUMENT PREPARATION SYSTEM

CLAMPS, HARDWARE

TO ALL HOLDERS OF DOD-STD-35-114(MI):

1. THE FOLLOWING PAGES OF DOD-STD-35-114(MI) HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED.

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
v	18 SEPTEMBER 1992	v	2 JUNE 1978
3	18 SEPTEMBER 1992	3	2 JUNE 1978
4	18 SEPTEMBER 1992	4	2 JUNE 1978
5	2 JUNE 1978	(Reprinted without change)	
6	18 SEPTEMBER 1992	6	2 JUNE 1978
7	18 SEPTEMBER 1992	7	2 JUNE 1978
8	18 SEPTEMBER 1992	8	2 JUNE 1978
15	18 SEPTEMBER 1992	15	2 JUNE 1978
16	18 SEPTEMBER 1992	16	2 JUNE 1978
35	2 JUNE 1978	(Reprinted without change)	
36	18 SEPTEMBER 1992	36	2 JUNE 1978
37	18 SEPTEMBER 1992	37	2 JUNE 1978
38	2 JUNE 1978	(Reprinted without change)	
59	2 JUNE 1978	(Reprinted without chanage	
60	18 SEPTEMBER 1992	60	2 JUNE 1978
61	18 SEPTEMBER 1992	61	2 JUNE 1978
62	18 SEPTEMBER 1992	62	2 JUNE 1978

2. RETAIN THIS NOTICE PAGE AND INSERT BEFORE THE TABLE OF CONTENTS.

3. Holders of DOD-STD-35-114(MI) will verify that page changes and additions indicated above have been entered. The notice page will be retained as a check sheet. This issuance, together with appended pages is a separate publication. Each notice is to be retained by stocking points until the Military Standard is completely revised or canceled.

Custodians:
Army-MI

Review activities:
Army-AT, AV, CR, ER
ME, MR, SC

User activity
Army - AR

Preparing activity:
Army - MI

(Project EGDS-A452)

AMSC N/A

AREA EGDS

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Downloaded from https://assistdvd.dla.mil/ on 2026-02-23T21:03Z
Check the source to verify that this is the current version before use.

CONTENTS

			Page
Paragraph	1.	SCOPE	1
	1.1	General	1
	1.2	Purpose	1
	1.3	Utilization	1
	1.4	Military Specification Exception Title Index	1
	2.	APPLICABLE DOCUMENTS	7
	2.1	Government documents	7
	2.2	Non-Government publications	8
	3.	DEFINITIONS	11
	4.	GENERAL REQUIREMENTS	13
	4.1	Introduction	13
	4.2	Specification Requirement Sheet (SRS)	13
	4.3	Configuration	13
	4.3.1	Dimensions	13
	4.3.2	Illustrations	13
	4.4	Units of measure	13
	5	DETAILED REQUIREMENTS	15
	5.1	Introduction	15
	5.2	MSE Formats	15
	5.3	Referenced documents	15
	5.4	Deleted	15
	5.5	Tabulation	16
	5.6	Keyword Code Index (KCI) and Standard Paragraphs	19
	5.6.1	Interrelationship of Codes	19
	5.6.2	Automatic MSE Data	19
	5.6.3	DOD-STD-35 Tables	19

TABLES

Table	I.	Title Index	3
	II.	Deleted	16
	III.	Tabulation Candidates	17

APPENDICES

Appendix	A.	Illustrations	79
	B.	Characteristic Code Index	91
		Index	95

Source: https://assist.dla.mil -- Downloaded: 202640-23T21:03Z
Check the source to verify that this is the current version before use.

TABLE I. TITLE INDEX

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
CLAMP, HALF	147001	MS39294	HYDRAULIC, 4-BOLT SPLIT FLANGE TYPE, 0 TO 4000 PSI, PRESSURE AND SUCTION LINE
CLAMP, HOSE AND TUBE	147002	WW-C-440	LOW PRESSURE
		MS35842	LOW PRESSURE, TYPE F (BAND)
		MS39326	SPRING, LOW PRESSURE, TYPE E (SPRING)
		MIL-C-6985	FLAT BAND
		AN737	AIRCRAFT
		MIL-C-8783	FLAT BAND 500 DEG.F
		MS21920	FLAT BAND 500 DEG.F
		MS22064	OXYGEN

Source: https://assist.dla.mil -- Downloaded: 202640-23T21:03Z
Check the source to verify that this is the current version before use.

TABLE I. TITLE INDEX (CONT)

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
CLAMP, HOSE AND TUBE	147002 (CONT)	MS53041	EXHAUST MUFFLER AND TAILPIPE
CLAMP, INSTRUMENT MOUNTING	147003	MIL-C-6818	AIRCRAFT
		MS28042	AIRCRAFT
CLAMP, LEAK REPAIR, PIPE COUPLING	147004	MIL-C-11418	METALLIC
CLAMP, LOOP	147005	MIL-C-8603	TUBE SUPPORT
		AN742	PLAIN
		MS21919	CUSHIONED
		MS21104	CUSHIONED 212 DEG. F
		MS21105	CUSHIONED, CRES 500 DEG. F
		MS21106	CUSHIONED, CRES 1200 DEG. F
		MS21122	CUSHIONED, PTFE
		AN735	BONDING

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:03Z
Check the source to verify that this is the current version before use.

TABLE I. TITLE INDEX (CONT)

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
CLAMP, LOOP	147005 (CONT)	MS9024	CUSHIONED, JOGGLED
		MS9025	SUPPORT
		MS9349	CUSHIONED, JOGGLED, CRES
		MS9350	CUSHIONED, JOGGLED, CRES
		MS9351	CUSHIONED, JOGGLED, ALUMINUM
		MS9352	CUSHIONED, JOGGLED, ALUMINUM
		MS9353	ALUMINUM
		MS9354	ALUMINUM
		MS9391	MULTIPLE TUBE
		MS21322	PLAIN AND CUSHIONED, ALUMINUM
		MS21333	PLAIND AND CUSHIONED, STEEL
		MS21334	PLAIN AND CUSHIONED, DOUBLE, STEEL
		MS25281	WIRE SUPPORT, PLASTIC

Source: https://assist.dla.mil -- Downloaded: 202640-23T21:03Z
Check the source to verify that this is the currentversion before use.

TABLE I. TITLE INDEX (CONT)

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
CLAMP, LOOP	147005 (CONT)	MS27405	PLATED OR UNPLATED, METALLIC
		MS122900 THRU MS122939	SUPPORT, HEAT RESISTANT
CLAMP, POLE LINE	147006	MS39032	CABLE SUSPENSION
		MS39041	ELECTRICIAL, STRAND, GROUND
		MS39046	ELECTRICAL, LASHING
CLAMP, REPAIR PIPE	147007	MIL-C-12386	SPLIT OR PIN HOLE, GASOLINE RESISTANT SEAL
CLAMP, WIRE ROPE	147008	FF-C-450	BOLTED AND THREADED
		MS16842	SINGLE SADDLE AND GRIP, STEEL
		MS16843	DOUBLE GRIP, THREADED, STEEL
		MS51868	DOUBLE SADDLE AND GRIP, STEEL

Source: https://assist.dla.mil -- Downloaded: 202610-23T21:03Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

FF-C-450	Clamps, Wire Rope
WW-C-440	Clamp Hose, (Low Pressure)

MILITARY

MIL-C-6985	Clamp, Hose
MIL-C-11418	Clamps, Leak Repair Pipe Coupling: Steel and Ductile Iron for Use Over Ground Type Couplings
MIL-C-12386	Clamp, Repair, Pipe; With Gasoline Resistant Gasket, General Specification For

STANDARDS

FEDERAL

FED-STD-595	Colors
FED-STD-601	Rubber: Sampling and Testing

MILITARY

MIL-STD-810	Environmental Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:03Z
Check the source to verify that this is the current version before use.

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

ASTM B 117	Standard Method of Salt Spray (Fog) Testing
ASTM D 395	Compression Set of Vulcanized Rubber
ASTM D 412	Tension Testing of Vulcanized Rubber
ASTM D 471	Standard Method of Test for Change in Properties of Elastomeric Vulcanizates Resulting from Immersion in Liquids
ASTM D 573	Accelerated Aging of Vulcanized Rubber by the Oven Method
ASTM D 624	Tear Resistance of Vulcanized Rubber
ASTM D 792	Standard Methods of Test for Specific Gravity and Density of Plastics by Displacement
ASTM D 1053	Standard Method for Measuring Low-temperature Stiffening of Rubber and Rubber-like Materials by Means of a Torsional Wire Apparatus
ASTM D 1149	Standard Method of Test for Accelerated Ozone Cracking of Vulcanized Rubber
ASTM D 2137	Standard Low-temperature Impact Test for Brittleness Determination of Flexible Polymeric Materials or Fabrics Coated, Therewith, or Both
ASTM D 2240	Standard Method of Test for Indentation Hardness of Rubber and Plastics by Means of a Durometer

Source: https://assist.dla.mil -- Downloaded: 202640-23T21:03Z
Check the source to verify that this is the currenversion before use.

5. DETAILED REQUIREMENTS

5.1 Introduction. This section provides instructions for completion of the SRS and references for instructions on specific requirements contained in DOD-STD-35.

5.2 MSE formats. AEDPS has the capability to prepare a six-section or a single-section document. The six-section document appears similar to a six-section military specification. The single-section document appears similar to a military specification sheet. Specification requirements described in DOD-STD-35 are applicable to either of the formats selected. It is mandatory that one of the two following codes and an "X" be entered on the SRS to create the desired MSE format:

a. 0700 for a six-section format

b. 0800 for a single-section format

5.3 Referenced documents. Section 2 of the MSE may contain a list of all referenced documents within Sections 3, 4, and 5 of the MSE. Instructions for referencing documents to appear in Section 2 of the forthcoming MSE are contained in DOD-STD-35.

5.4 Deleted

Source: https://assist.dla.mil -- Downloaded: 202610-23T21:03Z
Check the source to verify that this is the current version before use.

5.5 Tabulation. Instructions for tabulation are provided in DOD-STD-35. Table III provides codes for column headings which may be used for hardware clamps characteristics. When different units of measure within one system (U.S. or SI) are available for specifying a characteristic, only one of the units may be tabulated; e.g. either ounces or pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:03Z
Check the source to verify that this is the current version before use.

CHAR CODE	STANDARD PARAGRAPH
7022	MAXIMUM WEIGHT PER UNIT. THE WEIGHT PER UNIT SHALL BE NOT GREATER THAN 7022 OUNCE(S).
7023	MAXIMUM WEIGHT PER UNIT. THE WEIGHT PER UNIT SHALL BE NOT GREATER THAN 7023 POUND(S).
7021	MAXIMUM WEIGHT PER UNIT. THE WEIGHT PER UNIT SHALL BE NOT GREATER THAN 7021 GRAM(S).
7024	MAXIMUM WEIGHT PER UNIT. THE WEIGHT PER UNIT SHALL BE NOT GREATEER THAN 7024 KILOGRAM(S).
	PERFORMANCE.
	INSIDE DIAMETER ADJUSTMENT. THE INSIDE DIAMETER ADJUSTMENT SHALL BE AS FOLLOWS:
1630	THE INSIDE DIAMETER SHALL BE NOT LESS THAN 1630 INCHES.
1632	THE INSIDE DIAMETER SHALL BE NOT GREATER THAN 1632 INCHES.
1634	THE INSIDE DIAMETER SHALL BE NOT LESS THAN 1634 MILLIMETERS.
1636	THE INSIDE DIAMETER SHALL BE NO GREATER THAN 1636 MILLIMETERS.

Source: https://assist.dla.mil -- Downloaded: 202640-23T21:03Z
Check the source to verify that this is the currentversion before use.

KEYWORD CODE INDEX (KCI)

CHARACTERISTIC NAME	CHAR CODE
CLOSURE	
REQUIREMENTS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	1802
AS FOLLOWS:	1804
HOLES ALIGNMENT DEVIATION, MAXIMUM (SELECT ONE)	
(INCHES)	1806
(MILLIMETERS)	1808
CLAMP WIDTH AND CIRCUMFERENCE FLATNESS DEVIATION, MAXIMUM	
(INCHES)	1810
(MILLIMETERS)	1812
TEST METHODS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	1822
PER (DOC DES)	1824

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:03Z
Check the source to verify that this is the current version before use.

CHAR CODE	STANDARD PARAGRAPH
1802	CLOSURE. THE REQUIREMENT(S) SHALL BE IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 1802.
1804	CLOSURE. THE REQUIREMENT(S) SHALL BE AS FOLLOWS:
1806	THE CLAMP HOLES ALIGNMENT DEVIATION SHALL BE NOT GREATER THAN 1806 INCH(S).
1808	THE CLAMP HOLES ALIGNMENT DEVIATION SHALL BE NOT GREATER THAN 1808 MILLIMETER(S).
1810	THE CLAMP WIDTH AND CIRCUMFERENCE DEVIATION FROM FLATNESS SHALL BE NOT GREATER THAN 1810 INCH(S).
1812	THE CLAMP WIDTH AND CIRCUMFERENCE DEVIATION FROM FLATNESS SHALL BE NOT GREATER THAN 1812 MILLIMETER(S).
1822	CLOSURE. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 1822.
1824	CLOSURE. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH 1824.

Source: https://assist.dla.mil -- Downloaded: 202640-23T21:03Z
Check the source to verify that this is the currenversion before use.

KEYWORD CODE INDEX (KCI)

CHARACTERISTIC NAME	CHAR CODE
CUSHION OR GASKET	
REQUIREMENTS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	1902
AS FOLLOWS:	1904
NO FRACTURES, CRACKS, BREAKING OR SWELLING OF MATERIAL.	1906
MAXIMUM COMPRESSION SET DIMENSIONAL CHANGE (PERCENT)	1908
MINIMUM HARDNESS, DUROMETER (VALUE)	1910
MINIMUM SPECIFIC GRAVITY (NUMBER)	1912
MAXIMUM SPECIFIC GRAVITY (NUMBER)	1914
MINIMUM TEAR RESISTANCE (SELECT ONE)	
(KILONEWTONS/METER)	1916
(POUNDS/INCH)	1918
MINIMUM TENSILE STRENGTH (SELECT ONE)	
(POUNDS PER SQUARE INCH)	1920
(KILOPASCALS)	1922

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:03Z
Check the source to verify that this is the current version before use.

CHAR CODE	STANDARD PARAGRAPH
2622	LOAD RESISTANCE. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 2622.
2624	LOAD RESISTANCE. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH MIL-C-6985.
2626	THE HYDRAULIC GAGE PRESSURE SHALL BE NOT GREATER THAN 2626 POUNDS PER SQUARE INCH.
2628	THE HYDRAULIC GAGE PRESSURE SHALL BE NOT GREATER THAN 2628 KILOPASCALS.
2630	THE SCREW TORQUE SHALL BE NOT LESS THAN 2630 POUND-INCHES.
2632	THE SCREW TORQUE SHALL BE NOT LESS THAN 2632 NEWTON METERS.
2702	SAFE WORKING LOAD. THE REQUIREMENT(S) SHALL BE IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 2702.
2704	SAFE WORKING LOAD. THE REQUIREMENT(S) SHALL BE AS FOLLOWS:
2706	THE CLAMP GRIP SHALL BE NOT LESS THAN 2706 PERCENT OF THE WIRE ROPE BREAKING STRENGTH.
2708	THERE SHALL BE NO SEPERATION OF THE HOSE, CLAMP, OR COUPLING AS A RESULT OF THE VERTICAL PULL.

Source: https://assist.dla.mil -- Downloaded: 202610-23T21:03Z
Check the source to verify that this is the current version before use.

KEYWORD CODE INDEX (KCI)

CHARACTERISTIC NAME	CHAR CODE
SAFE WORKING LOAD (CONT)	
TEST METHODS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	2722
PER FF-C-450.	2724
WIRE ROPE BREAKING STRENGTH (SELECT ONE)	
(POUNDS)	2726
(KILOGRAMS)	2728
PER (DOC DES)	2730
MINIMUM VERTICAL PULL (SELECT ONE)	
(POUNDS)	2732
(NEWTONS)	2734
DURATION (MINUTES)	2736
SALT SPRAY/FOG	
REQUIREMENTS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	6356
NO EVIDENCE OF CORROSION, PHYSICAL DAMAGE, OR DEGRADATION OF PERFORMANCE.	6357

Source: https://assist.dla.mil -- Downloaded: 202640-23T21:03Z
Check the source to verify that this is the currenversion before use.

CHAR CODE	STANDARD PARAGRAPH
2722	SAFE WORKING LOAD. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 2722.
2724	SAFE WORKING LOAD. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH FF-C-450.
2726	THE BREAKING STRENGTH OF THE ROPE SHALL BE 2726 POUNDS.
2728	THE BREAKING STRENGTH OF THE WIRE ROPE SHALL BE 2728 KILOGRAMS.
2730	SAFE WORKING LOAD. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH 2730.
2732	THE VERTICAL PULL SHALL BE NOT LESS THAN 2732 POUNDS.
2734	THE VERTICAL PULL SHALL BE NOT LESS THAN 2734 NEWTONS.
2736	THE DURATION OF VERTICAL PULL SHALL BE 2736 MINUTES.
6356	SALT SPRAY/FOG. THE REQUIREMENT(S) SHALL BE IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 6356.
6357	SALT SPRAY/FOG. THERE SHALL BE NO EVIDENCE OF CORROSION, PHYSICAL DAMAGE, OR DEGRADATION OF PERFORMANCE.

Source: https://assist.dla.mil - Downloaded: 202610-23T21:03Z
Check the source to verify that this is the current version before use.

KEYWORD CODE INDEX (KCI)

CHARACTERISTIC NAME	CHAR CODE
SALT SPRAY/FOG (CONT)	
TEST METHODS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	6353
PER ASTM B 117.	6354
EXPOSURE PERIOD (HOURS)	4783
TENSILE STRENGTH	
REQUIREMENTS (SELECT ONE)	
PER BASE DOCUMENT ____	2802
AS FOLLOWS:	2804
MINIMUM TENSILE STRENGTH (SELECT ONE)	
(POUNDS PER SQUARE INCH)	2806
(KILOPASCALS)	2808
TEST METHODS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	2822
PER ASTM E 8.	2824
MAXIMUM LOAD (SELECT ONE)	
(POUNDS)	2826
(KILOGRAMS)	2828

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:03Z
Check the source to verify that this is the current version before use.